



Answer: A

30 Actual count-rate of the source, $C_0 = 100 - 20 = 80 \text{ min}^{-1}$

After 12 hours, actual count-rate, $C = C_0 \left(\frac{1}{2} \right)^{\frac{t}{t_{1/2}}} = 80 \left(\frac{1}{2} \right)^{\frac{12}{48}} \approx 67 \text{ min}^{-1}$

Observed count-rate = $67 + 20 = 87 \text{ min}^{-1}$ (Need to add background radiation)

Answer: C